
DIPTIMAN BORA

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EDUCATION

The University of Arizona

Bachelor of Science in Physics (*with Honors*) & Data Science (*Computing Emphasis*)

Minor in Computer Science and Mathematics

Honors: Global Wildcat Award (\$23,000 per year), US-ATLAS ATC Award

Member, W.A. Franke Honors College

Tucson, AZ

Expected May 2028

GPA: 3.946/4.00

RESEARCH EXPERIENCE

Brookhaven National Laboratory, Omega Group

May 2026 - present

Research Intern (Advisor: Dr. Ang Li)

- Performed rigorous quality control (QC) testing such as IV scans, gain and noise plots, on ATLAS Inner Tracker (ITk) Strip Barrel modules sourced from SCIPP, LBNL, and BNL for Inner Detector upgrade in HL-LHC upgrade
- Engineered a coolant-leakage monitoring system measuring gas resistance, humidity and temperature using a metal-oxide sensor and Raspberry Pi in ITk stave testing enclosure
- Developed and maintained the “BNL Uploader,” a custom automation tool to streamline QC results into a database

Department of Physics, The University of Arizona

October 2025 - present

Undergraduate Research Intern (Advisor: Dr. Nikolay Golubev)

- Applied theoretical frameworks in molecular and ultrafast dynamics, including Born–Oppenheimer approximation and the exact factorization of the molecular wavefunction, toward future modeling work

Department of Physics, The University of Arizona

October 2024 - present

Undergraduate Research Intern (Advisor: Dr. Kenneth Johns)

- Developed more than 10 Python scripts to test FPGA firmware being developed for ATLAS Liquid Argon calorimeter readout at CERN LHC, generating test data for longer-duration testing of these boards via SSH
- Applied ML models (MLPs, Graph Neural Networks) to differentiate long-lived particles (LLPs) in ATLAS Muon Spectrometer Run 2 data, optimizing GPU-based training on Monte Carlo and real datasets, and now preparing for new Run 3 data

PRESENTATIONS

Studying Coolant Leakage in Stave Testing Box

July 2026

US ATLAS ITk Strips Technical Meeting

[July 10 Presentation at Indico](#)

[July 24 Presentation at Indico](#)

Identifying Long Lived Particles with ML

June 2025

ATLAS MS Vertex Group

AWARDS & HONORS

US-ATLAS ATC Award

2026

- Awarded funding to support research with the ATLAS experiment at Brookhaven National Laboratory.

George Gregson Scholarship in Physics

2026

Galileo Circle Scholar

2026

Dean’s List with Distinction

Spring 2025, Fall 2025, Spring 2026

Dean’s List

Fall 2024

Global Wildcat Award

Fall 2024 - present

ACADEMIC PROJECTS

Dipole Field Trajectory Simulator

January 2026 - present

- Created an iterative numerical code to model the trajectory of a charged particle moving in a dipole field at the origin

Global Food Insecurity ETL Pipeline

August 2025 - December 2025

- Engineered an interactive web dashboard using Streamlit to integrate multiple data sources and analyze the variation of food insecurity across countries, and its effect on income and population density
- Developed an ETL pipeline to ingest and normalize large-scale demographic data via API requests and CSVs into a relational SQL database

TEACHING EXPERIENCE

Content Tutor for THINK TANK (Physics)

The University of Arizona

2026 - 2027

- Provide academic assistance and subject mastery tutoring to students in group and one-on-one settings

Preceptor: PHYS 111: Introductory Studio Physics 2

Department of Physics, The University of Arizona

Spring 2026

- Collaborated with TAs in weekly lab sessions to troubleshoot experiments, demonstrate physical concepts, and resolve student doubts

Preceptor: PHYS 241: Introductory Electricity and Magnetism

Department of Physics, The University of Arizona

Fall 2025

- Guided 250+ students during group work alongside a team of preceptors and TAs

SKILLS, CERTIFICATIONS & OTHER

Programming & Software:	Python, SQL, Java, Git, Linux, Windows, Jupyter, Streamlit, Grafana, $\text{\LaTeX}$
Machine Learning & Data:	Data Pipeline Implementation, Data Analysis, Data Visualization, Deep Learning
Hardware:	Raspberry Pi, Arduino, FPGA testing, BBC micro:bit, general electronics
Languages:	English (Professional), Assamese (Native), Hindi (Native)
Certifications:	Stanford University & DeepLearning.AI Machine Learning Specialization (Coursera, May 2025), Data Analysis with Python by IBM (IBM, March 2025), CS50's Introduction to Programming with Python (HarvardX, Dec. 2023), 3D Printing Certification – University of Arizona
Hobbies:	Football (soccer), racketball, pool, classical guitar, running, singing